

Scientific Reviewer

13+3 · Accept with notes

Clean inventum · fair foils · ablations · physics suite · candid limits. Ask:
Fig.6 table · “first” boundaries · CEM budget · TwoRoom fix · real-robot ·
cite 2-AC.

SPINE: Stable E2E JEPA WM · MSE+SIGReg only · ~15M · 48× vs DINO-WM (0.98s/47s) · SIBLING to Meta JEPA · SIM-ONLY · PushT 96 ± 2.83 ·
fixed-FLOP 90vs13 · recon hurts · λ -only

Claims under review

What the paper asserts

C1 First stable E2E JEPA from raw pixels with two-term MSE+SIGReg

Strong — check boundaries

C2 Removes EMA/SG/freeze/multi-VICReg; λ -only effective HP

Well supported vs PLDM

C3 $\sim 15\text{M}$ · $\sim 48\times$ planning vs DINO-WM · competitive sim control

Fig.3/6/Tab.5

C4 Latent encodes physics (probe + VoE + straightening)

Beyond reward hacks

Strengths

Why this clears the bar

Clean inventum

Collapse problem → two-term objective → measurable stability + planning speed.

Fair foils

E2E (PLDM) · foundation (DINO-WM) · GC-RL/BC — spans the taxonomy.

Ablations isolate thesis

Recon hurts; λ robust band; M inert; dropout 0.1; predictor size sweet spot.

Physics suite

Probe + VoE (teleport $p < 0.01$) + emergent temporal straightening.

Candid limits

Short horizon; SIGReg vs low-dim; action labels — stated, not hidden.

Open stack

Code + site claimed; Alg.3 reproducible.

Ablations — Fig. 15 / 16

Emb dim · M · knots · λ — λ is the only real knob

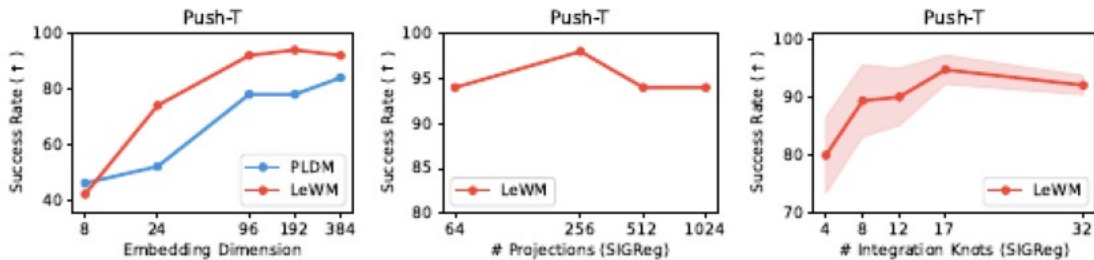


Figure 15: **Ablation studies of key design choices in LeWM.** **Left:** effect of the embedding dimension; performance improves with larger embeddings but quickly saturates beyond a certain threshold. **Center:** effect of the number of random projections used in SIGReg; performance remains stable, indicating that this parameter is not critical. **Right:** effect of the number of integration knots used to compute the SIGReg loss; results are similarly insensitive to this parameter.

Table 5: **Training Variance.** We report the mean success rate across three training seeds and the corresponding variance, evaluated over the same set of 50 trajectories on Push-T. The goal configuration is reachable within 25 steps, and we allow a planning budget of 50 steps. PLDM exhibits higher variance compared to DINO-WM and LeWM.

Model	Push-T (SR $\uparrow$)
DINO-WM	92.0 $\pm$ 1.63
PLDM	78.0 $\pm$ 5.0
LeWM (ours)	96.0 $\pm$ 2.83

Embedding dimensions. We study the impact of the embedding dimensionality on performance. As shown in Fig. 15, performance drops when the embedding dimension falls below a certain threshold (around 184), while increasing the dimension beyond this value yields diminishing returns and leads to performance saturation.

Number of projections in SIGReg. We study the impact of the number of projections used in

Pair with Tab.5-10. Demand numeric table for Fig.6 non-PushT envs.

Physics suite — VoE Fig. 8

Teleport surprise $p < 0.01$ · color change weaker · physical > visual

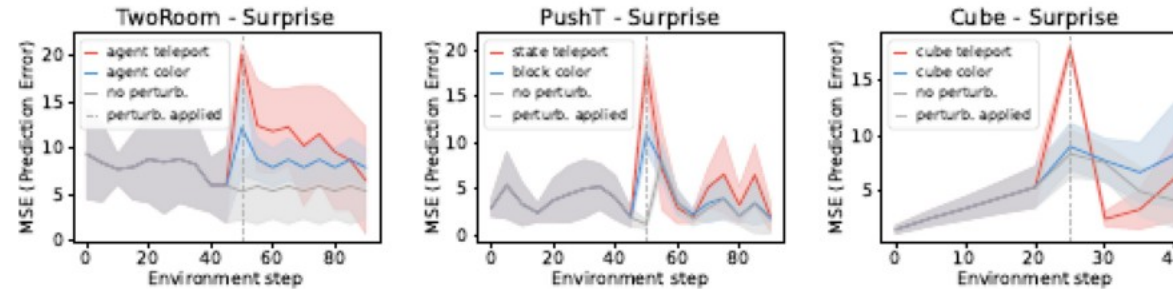


Figure 8: **Violation-of-expectation evaluation across three environments.** Each plot shows LeWM surprise along three trajectories: an unperturbed reference, a visually perturbed trajectory with abrupt object color change, and a physically perturbed trajectory where objects teleport to random positions. Teleportation violates physical continuity and produces a pronounced surprise spike, while the unperturbed trajectory stays at a low baseline. The increase is significant for teleportation across all environments (paired t-test, $p < 0.01$) but weaker and non-significant for color changes, indicating greater sensitivity to physical than visual perturbations. Environments, left to right: *TwoRoom*, *PushT*, *OGBench Cube*.

increasingly straight on PushT over training as a purely emergent phenomenon, without any explicit regularization encouraging this behavior, cf. Fig. 17. Remarkably, LeWM achieves higher temporal straightness than PLDM, despite PLDM employing a dedicated temporal smoothness regularization term. We detail our findings in App. H.

5.2 Violation-of-expectation Framework

Another approach to quantifying physical understanding is the ability to detect violations of the learned world model. Inspired by the violation-of-expectation (VoE) paradigm used in developmental psychology and recently adopted in machine learning [46–48], this framework evaluates whether a

Strength beyond SR bars — world-model claim needs this suite.

Weaknesses / ask-for-revision

Notes that should travel with accept

Fig.6 chart-heavy

Push-T Tab.5 good; other envs lean on figure reading — request full numeric table.

“First” claim

Depends on definitional boundaries vs LeJEPa / Causal-JEPa concurrent work.

CEM budget

300×30 dominates wall-clock; matched-iteration budgets beyond Fig.3.

TwoRoom fix

Diagnosis $\neq$ cure — want adaptive λ /dim/covariance experiment.

No real-robot

LeCun-coauthored WM → reviewers ask Franka/Bridge smoke.

Decoder viz

Open-loop figs use post-hoc decoder — clarify not training signal.

Citation rules — do not launder robotics

CLEAR vs wrong cite

When citing real-robot JEPA planning → V-JEPA 2-AC

Franka / DROID / zero-shot robot SR are NOT in LeWM. Paper is sim Push-T / OGBench / TwoRoom / Reacher only.

When citing stable E2E recipe → LeWM / LeJEPA SIGReg

Two-term MSE+SIGReg · λ -only · no EMA/SG/freeze — this paper's inventum for the small control-WM niche.

Red flag: attributing Franka success to 2603.19312 alone.

Internal reuse checklist

Before citing in club / repo

- ☐ Reproduce Tab.5 Push-T 3-seed SR with lucas-maes/le-wm
- ☐ Matched-FLOP Fig.3 protocol vs DINO-WM on our GPU
- ☐ λ bisection on a new env (another OGBench task)
- ☐ Real-robot JEPA planning → cite V-JEPA 2-AC, not LeWM
- ☐ Stable E2E recipe → cite LeWM / LeJEPA SIGReg
- ☐ Clarify decoder figs are post-hoc viz, not train loss
- ☐ Do not invent within-LeWM EMA/SG factorial as paper result
- ☐ Disclose TwoRoom underperformance + Cube DINO edge

Speed claim under review — Fig. 3

Accept directionally · ask matched wall-clock + matched CEM iters

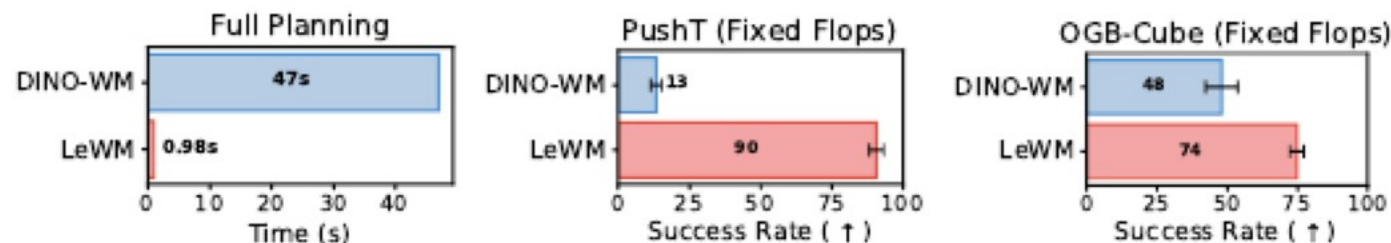


Figure 3: **Planning time and performance under fixed compute.** **Left:** Planning time comparison averaged over 50 runs. Encoding observations with $\sim 200\times$ fewer tokens than DINO-WM allows LeWM to achieve planning speeds comparable to PLDM while being up to $\sim 50\times$ faster than DINO-WM. **Center-Right:** Planning performance under the same computational budget (fixed FLOPs). LeWM significantly outperforms DINO-WM on Push-T (center) and OGBench-Cube (right). See App. D for planning setup details.

- We evaluate physical understanding in the latent space through probing of physical quantities and a violation-of-expectation test for detecting unphysical trajectories.

2 Related Work

World Models aim to learn predictive models of environment dynamics from data, enabling agents to reason about future states in imagination. A prominent class of WMs consists of *generative* approaches that explicitly model environment dynamics in pixel space. These action-conditioned generative models act as learned simulators by producing future observations conditioned on past

Speedup largely cheaper dynamics (compact z). Still a real systems contribution.

Verdict

Accept with notes

ACCEPT — landmark small-WM recipe

Clean two-term inventum, strong Push-T + speed evidence, candid limits, open code trajectory.

WITH NOTES

Numeric table for Fig.6 · soften “first” · CEM matched budgets · TwoRoom adaptive fix · real-robot smoke or explicit deferral to 2-AC · decoder-viz clarification.

Do not reject for missing Franka — but require cite-rule discipline in follow-ons.

Exam bite — Reviewer

30-second critical summary

LeWM earns accept-with-notes: the two-term E2E JEPA-WM inventum is unusually clean, foils are fair, and Push-T/speed/ablations back the stability story. Demand a full SR table, matched CEM compute, and a TwoRoom fix path. Never let readers confuse this sim recipe with V-JEPA 2-AC Franka results — cite 2-AC for robots, LeWM for λ -only SIGReg E2E.

Numbers tattoo — pin these

Shared spine across all five seats

~15M

params · 1 GPU / few h

0.98s vs 47s

~48× planning speed

~200×

fewer tokens vs DINO

90 vs 13

fixed-FLOP Push-T

74 vs 48

fixed-FLOP Cube

96.0±2.83

Push-T SR (Tab.5)

78±5 / 92±1.63

PLDM / DINO Push-T

$\lambda \in [0.01, 0.2]$

SR>80%; peak ~0.09

96 → 86

recon loss hurts (T7)

SIBLING

not Meta descendant

SIM-ONLY

cite 2-AC for robots

λ -only

vs PLDM 6 knobs

Scientific Reviewer — close

Takeaways + exam bite

1. Strengths: clean inventum, fair foils, ablations, physics suite, candid limits.
2. Weaknesses: Fig.6 numeric table; “first” boundaries; CEM budget; TwoRoom; no robot.
3. Cite rule: robots → 2-AC; stable E2E recipe → LeWM/SIGReg.
4. Verdict: ACCEPT with notes — landmark small-WM recipe.

EXAM BITE

Accept-with-notes: clean two-term inventum; enforce cite-rule so Franka stays on 2-AC.